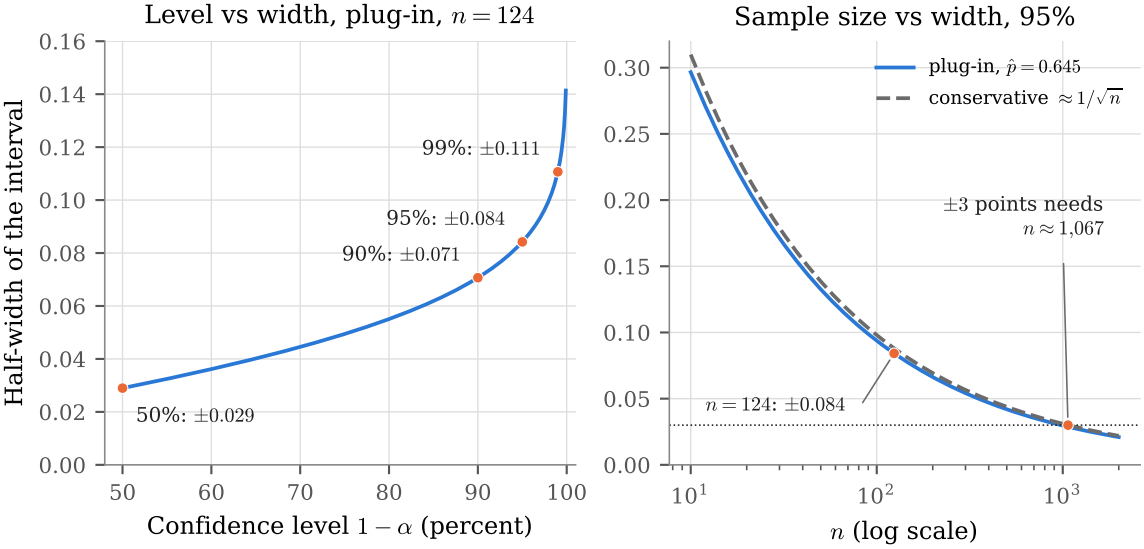
Level vs width, plug-in, $n = 124$
Half-width of the interval
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
99%: ±0.111
95%: ±0.084
90%: ±0.071
50%: ±0.029
50
60
70
80
90
100
Confidence level $1 - \alpha$ (percent)
Sample size vs width, 95%
plug-in, $\hat{p} = 0.645$
conservative $\approx 1/\sqrt{n}$
0.30
0.25
0.20
0.15
0.10
0.05
0.00
±3 points needs
$n \approx 1{,}067$
$n = 124$: ±0.084
10^1
10^2
10^3
n (log scale)